TRAVEL BUDGET ANALYSIS

Python Programming Case Study

1. Problem Statement

Planning a trip involves expenses in a foreign currency, such as food, hotel, transport, and shopping. Converting the total expenses into Indian Rupees and including a currency conversion fee can be difficult to calculate manually. A program is needed to calculate the total travel cost, identify the largest expense, and check whether the trip stays within the planned budget.

2. Objective

To develop a Python program that accepts the travel budget, exchange rate, conversion fee, and individual travel expenses; calculates the total foreign expense and converted INR amount; identifies the largest expense; and reports whether the final cost is within the budget.

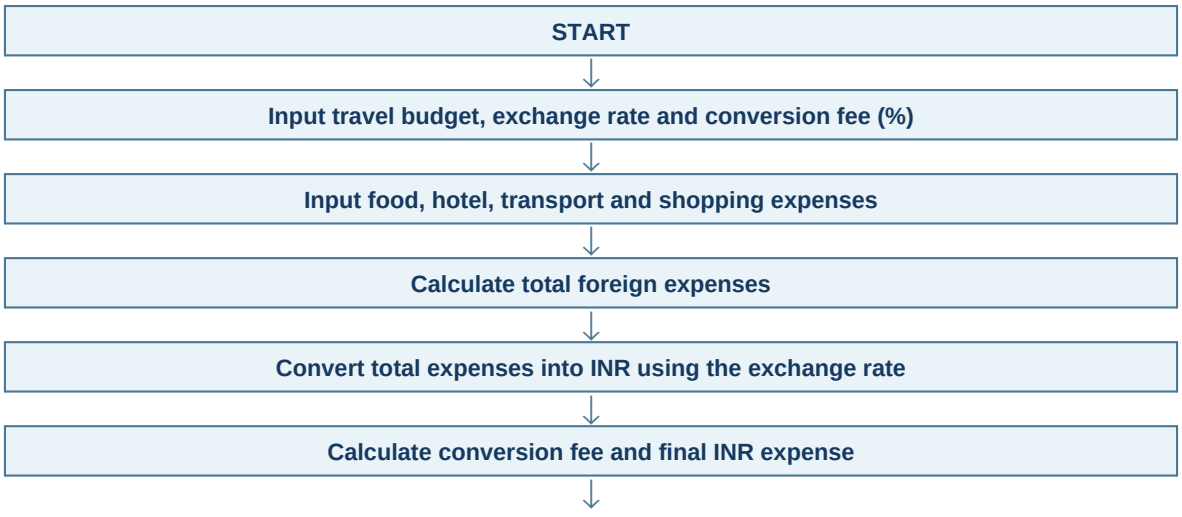
3. Assumptions

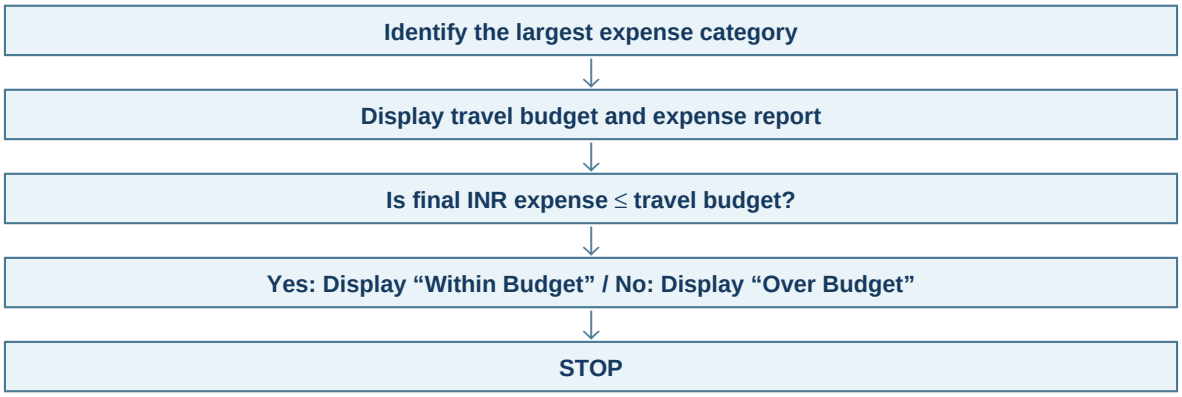
• All entered values are valid numeric values and expenses/budget are non-negative.
• Food, hotel, transport, and shopping expenses are entered in the same foreign currency.
• The exchange rate represents Indian Rupees for one unit of the foreign currency.
• The conversion fee is entered as a percentage and is calculated on the converted INR amount.
• Final INR expense equals converted amount plus the conversion fee.
• The largest expense is determined by comparing the four foreign-currency expense categories.
• The budget check compares the final INR expense with the entered INR budget.

4. Algorithm

1. Start the program.
2. Read the travel budget, exchange rate, and conversion fee percentage.
3. Read food, hotel, transport, and shopping expenses in foreign currency.
4. Add the four expenses to calculate the total foreign expense.
5. Multiply the total foreign expense by the exchange rate to get the converted INR amount.
6. Calculate the conversion fee as a percentage of the converted amount.
7. Add the conversion fee to obtain the final INR expense.
8. Find the expense category with the highest value.
9. Display the travel budget, expense totals, converted amount, fee, final INR expense, and largest expense.
10. If the final INR expense is less than or equal to the budget, display “Within Budget”; otherwise display “Over Budget”.
11. Stop the program.

5. Flowchart



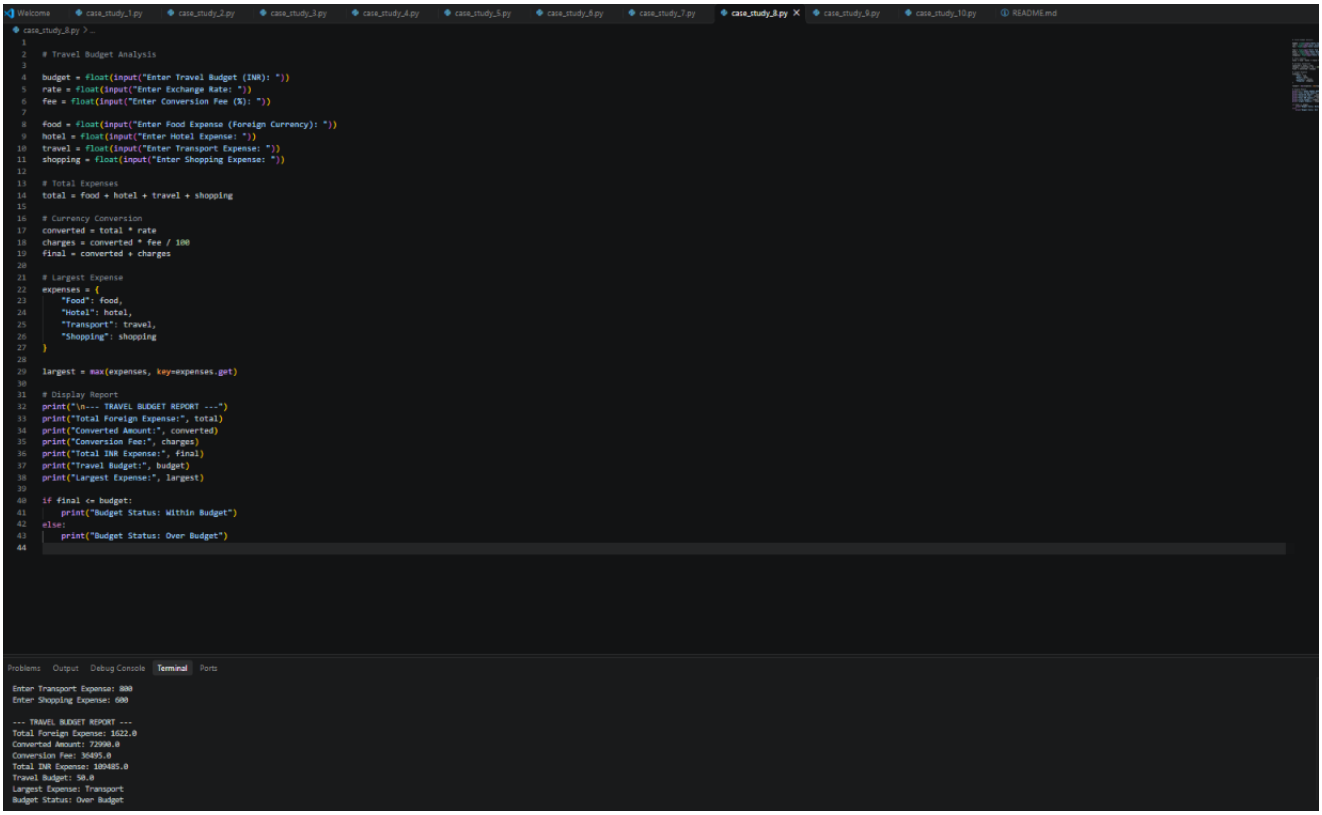

Identify the largest expense category
Display travel budget and expense report
Is final INR expense ≤ travel budget?
Yes: Display “Within Budget” / No: Display “Over Budget”
STOP

6. Pseudocode

```
BEGIN
INPUT budget, rate, fee
INPUT food, hotel, travel, shopping
total = food + hotel + travel + shopping
converted = total * rate
charges = converted * fee / 100
final = converted + charges
expenses = {Food: food, Hotel: hotel,
Transport: travel, Shopping: shopping}
largest = category with maximum expense value
DISPLAY total, converted, charges, final, budget, largest
IF final <= budget THEN
DISPLAY 'Budget Status: Within Budget'
ELSE
DISPLAY 'Budget Status: Over Budget'
END IF
END
```

7. Python Source Code and Output Screenshot

The original screenshot supplied for this case study is included on the next page.



```python
# Travel Budget Analysis

budget = float(input("Enter Travel Budget (INR): "))
rate = float(input("Enter Exchange Rate: "))
fee = float(input("Enter Conversion Fee (%): "))

food = float(input("Enter Food Expense (Foreign Currency): "))
hotel = float(input("Enter Hotel Expense: "))
travel = float(input("Enter Transport Expense: "))
shopping = float(input("Enter Shopping Expense: "))

# Total Expenses
total = food + hotel + travel + shopping

# Currency Conversion
converted = total * rate
charges = converted * fee / 100
final = converted + charges

# Largest Expense
expenses = {
    "Food": food,
    "Hotel": hotel,
    "Transport": travel,
    "Shopping": shopping
}

largest = max(expenses, key=expenses.get)

# Display Report
print("\n--- TRAVEL BUDGET REPORT ---")
print("Total Foreign Expense:", total)
print("Converted Amount:", converted)
print("Conversion Fee:", charges)
print("Total INR Expense:", final)
print("Travel Budget:", budget)
print("Largest Expense:", largest)

if final <= budget:
    print("Budget Status: Within Budget")
else:
    print("Budget Status: Over Budget")
```

```
Problems  Output  Debug Console  Terminal  Ports

Enter Transport Expense: 800
Enter Shopping Expense: 600

--- TRAVEL BUDGET REPORT ---
Total Foreign Expense: 1622.0
Converted Amount: 72990.0
Conversion Fee: 36495.0
Total INR Expense: 109485.0
Travel Budget: 50.0
Largest Expense: Transport
Budget Status: Over Budget
```

8. Conclusion

The Travel Budget Analysis program calculates travel expenses across multiple categories, converts the total into Indian Rupees, includes the conversion fee, identifies the largest expense, and checks the final amount against the planned budget. It demonstrates Python input, arithmetic operations, dictionaries, the max() function, and conditional statements in a practical travel-planning task.